



Master Guide | PUBLIC

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SAP Identity Management Master Guide

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1 SAP Identity Management Master Guide

The SAP Identity Management Master Guide gives an overview of its architecture and components and also contains a scenario overview and a description of the installation procedure.

Related Information

[Getting Started \[page 4\]](#)

[SAP Identity Management Overview \[page 8\]](#)

[Overall Implementation Sequence \[page 24\]](#)

[SAP Identity Management Scenarios \[page 27\]](#)

[Appendix \[page 43\]](#)

2 Getting Started

This section contains useful information about the document and references to other relevant documents.

Related Information

[About this Document \[page 4\]](#)

[Related Information \[page 6\]](#)

[Important SAP Notes \[page 7\]](#)

2.1 About this Document

This Master Guide is the central starting point for the technical implementation of SAP Identity Management. You can find cross-scenario implementation information as well as scenario-specific information in this guide.

The Master Guide provides an overview of SAP Identity Management, its software units, and its scenarios from a technical perspective. Use it to help you design your identity management system landscape before you start the implementation phase. It refers you to the required detailed documentation, mainly:

- Installation guides for individual software components
- SAP Notes
- Configuration documentation
- Tutorials

📘 Note

Update information is included in the *Identity Management Installation and Update Guide* on Windows and UNIX for the individual software components. The following documents are relevant.

Update and Upgrade Documents

Document	Description
Updating SAP Identity Management on Windows	This section in the <i>Identity Management Installation and Update Guide</i> describes how to update SAP Identity Management components to a higher SP level.
Updating SAP Identity Management on UNIX	

Document	Description
Upgrading SAP Identity Management from 7.2 to 8.0	This document describes how you use the provisioning framework for SAP systems to upgrade a production system from SAP NetWeaver Identity Management 7.2 to 8.0.

The Master Guide consists of the following main sections:

- **SAP Identity Management Overview**
This section provides an overview of SAP Identity Management, including its software components, the connectors and frameworks that are delivered, and information about the solution-wide capabilities that apply to all scenarios. It also provides an overview of the system landscape and the overall implementation sequence.
- **SAP Identity Management Scenarios**
This section provides an overview of the identity management scenarios:
 - Provisioning for SAP or non-SAP systems
 - Integration with SAP Human Capital Management (SAP HCM)
 - Enhanced SAP Business Suite integration
 - Integration with BusinessObjects Access Control
 - Federation

Note

You can implement any or all of the scenarios in your landscape.

Note

You can find the most current information about the technical implementation of SAP Identity Management and the latest installation and configuration guides at [SAP Identity Management 8.0](#).

We strongly recommend that you use the documents available here. The guides are regularly updated.

Constraints

- The business scenarios that are presented here serve as examples of how you can use SAP software in your company. The business scenarios are only intended as models and do not necessarily run the way they are described here in your customer-specific system landscape. Check your requirements and systems to determine whether these scenarios can be used productively at your site. Furthermore, we recommend that you test these scenarios thoroughly in your test systems to ensure they are complete and free of errors before going live.
- This Master Guide primarily discusses the overall technical implementation of SAP Identity Management, rather than its subordinate components. This means that additional software dependencies might exist without being mentioned explicitly in this document. You can find more information on component-specific software dependencies in the corresponding installation guides.
- Good data quality is a prerequisite for the successful implementation of an identity management system. Before you start implementing SAP Identity Management, we recommend that you clean up the identity data in the systems that you want to integrate.

2.2 Related Information

For more information about topics not covered in this guide, see the following content on SAP Help Portal:

Related Information

Content	Location on SAP Service Marketplace or SCN
Latest version of installation guide	SAP Identity Management Installation and Update Guide on Windows SAP Identity Management Installation and Update Guide on UNIX
General information about SAP Identity Management	SAP Identity Management Community 
Sizing, calculation of hardware requirements	Sizing Guide for SAP NetWeaver Identity Management 7.1/7.2
Released platforms and technology-related topics, such as maintenance strategies and language support	Product Availability Matrix  SAP on SQL Server Community  Other database and operating systems 
Network security	SAP Identity Management Security Guide
High Availability	SAP Identity Management Solution Operation Guide
Information about Support Package Stacks, latest software versions, and patch level requirements	SAP Support Portal 

Related Information

[Further Useful Links \[page 6\]](#)

2.2.1 Further Useful Links

The following table lists further useful links on SAP Service Marketplace.

Content	Location on SAP Service Marketplace
Information about creating error messages	Incidents 
SAP Notes Search	SAP Note & Knowledge Base Article Search 

Content	Location on SAP Service Marketplace
SAP Software Download Center (software download and ordering of software)	SAP Software Download Center 
SAP Online Knowledge Products (OKPs) – role-specific learning maps	Get Involved Early 

2.3 Important SAP Notes

Read the following SAP Notes before you start the installation.

These SAP Notes contain the most recent information about installation, as well as corrections to the installation documentation. Make sure that you have the up-to-date version of each SAP Note, which you can find on SAP Service Marketplace.

SAP Note [2036858](#) 

Related Information

[SAP Note & Knowledge Base Article Search](#) 

3 SAP Identity Management Overview

This section provides an overview of SAP Identity Management.

Related Information

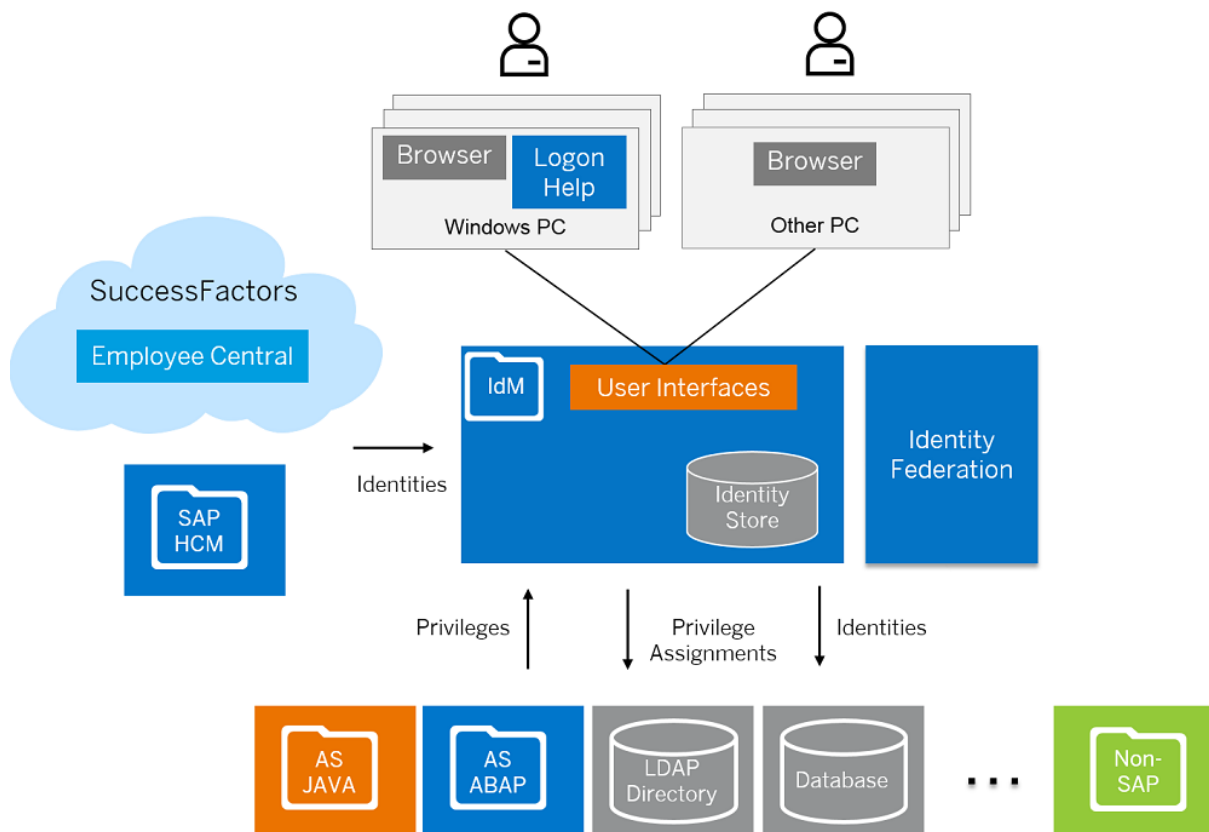
[Introduction to SAP Identity Management \[page 8\]](#)

[Software Components of SAP Identity Management \[page 11\]](#)

[Solution-Wide Capabilities \[page 22\]](#)

3.1 Introduction to SAP Identity Management

Enterprises are under pressure to increase the speed of deploying new applications and systems across their global networks, both internally and in the context of e-business with partners and customers. One of the challenges involved in these processes is the difficulty in finding and bringing together information relating to identities and resources that are distributed across multiple and often incompatible information sources. Identity data is often stored in many different applications throughout the enterprise and maintained manually in different locations. This is costly and, in addition to posing a security risk, can cause inconsistencies and low data quality. The prime objective of SAP Identity Management is to centrally manage and keep all identity data within the enterprise up-to-date.



This diagram illustrates a system landscape where SAP Identity Management is used to maintain identity data. Identities are imported to the Identity Management identity store from SAP HCM based on available authorizations.

Privileges are added from the back-end systems and identities are assigned to them. These privilege assignments are then written back to the back-end systems.

Identity Federation is a separate component that is used for Single Sign-On (SSO) for SAP and non-SAP systems. It supplies a SAML 2.0-compliant identity provider for Web-based access, and a security token service for Web services SSO.

End users use one of the Identity Management User Interface options to perform self-service or other tasks on the data in the identity store. If the client runs on Microsoft Windows, Logon Help is available to reset the password.

3.1.1 Identity Federation

Identity federation includes a SAML 2.0 identity provider and a security token service (STS) using the WS-Trust 1.3 standard.

You can use the identity provider for single sign-on (SSO) with SAP or non-SAP service providers. As an identity provider, SAP Netweaver Application Server (SAP NetWeaver AS) Java can provide cross-domain SSO in combination with SAML 2.0 service providers and at the same time enable single log-out (SLO) to close all user sessions in the SAML landscape. SAML 2.0 also enables identity federation by defining a name ID to be shared between the identity provider and one or more service providers.

You can use the STS to provide cross-domain SSO for web service providers. The STS converts what are often proprietary authentication methods from a Web service consumer into a security token consumable by the web service provider. The STS supports X.509, SAML 1.1, and SAML 2.0 security token types.

The identity federation component runs separately from the rest of SAP Single Sign-On. It can be installed together with the other components, but there are no technical dependencies between the identity federation component and the other SAP Single Sign-On components.

You can deploy this software on SAP NetWeaver AS for Java release 7.2 SPS 2 with SAP Note [1471322](#) applied or SAP NetWeaver AS for Java release 7.2 SPS 3 or later. However, to use the security token service or the newest user interface improvements in the identity provider, you must install the latest identity federation software component archive (SCA) and upgrade the host SAP NetWeaver AS for Java to release 7.2 SPS 4 or later.

Related Information

[Identity Provider for SAP Single Sign-On and SAP Identity Management](#)
[Security Token Service for SAP Single Sign-On and SAP Identity Management](#)

3.1.2 Logon Help

SAP Identity Management Logon Help is a client application for Microsoft Windows workstations that users can use to reset their passwords.

→ Recommendation

When installing SAP Identity Management 8.0, we recommend that you no longer install the Logon Help component. This is due to the fact that Logon Help is supported for Windows 7 which is out of maintenance. Existing customers, who are already using Logon Help component with Windows 7, will still be supported. For new installations and scenarios customers can use the Windows build-in alternative.

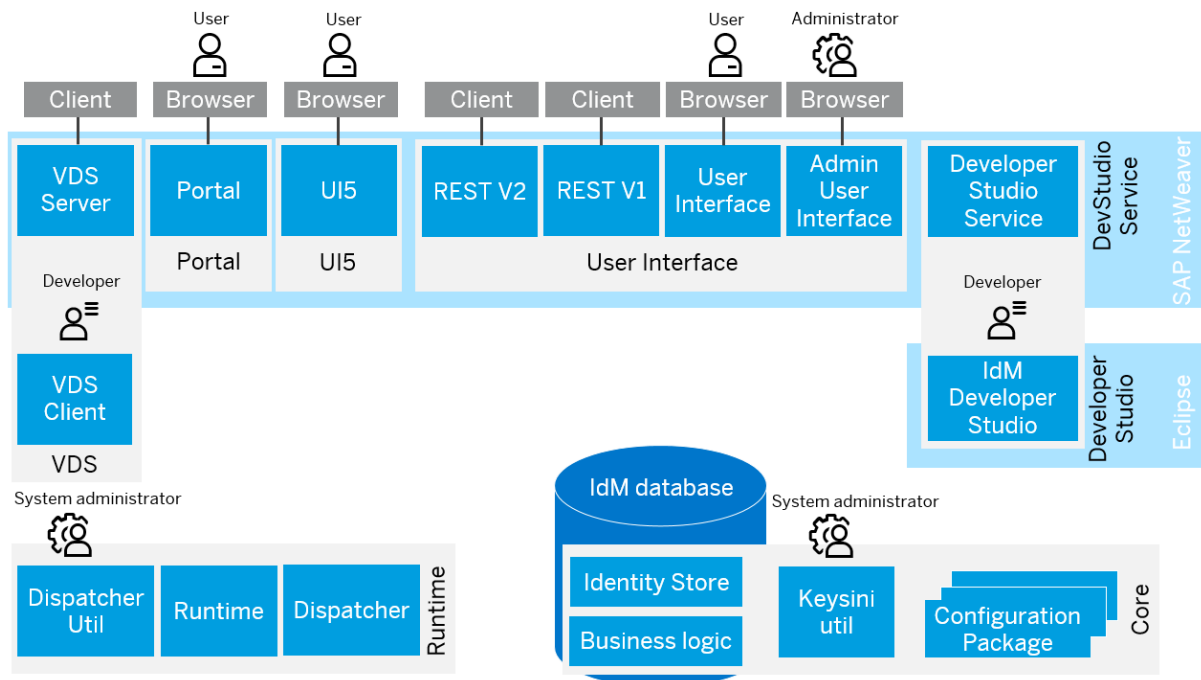
Logon help performs the password reset in conjunction with the Password Reset Self-Service scenario of SAP Identity Management Identity Center and a Microsoft Windows domain controller. Business users set their security questions and answers as part of the self-service scenario. If the business users forget their password for logging on to the Microsoft Windows domain on their workstation, business users can use the front-end client, Logon Help, to enter answers to security questions and a new password. If the business users enter their data correctly, Logon Help logs the business users on to the Microsoft Windows domain with the new password.

Related Information

[Logon Help for SAP Identity Management Implementation Guide](#)

3.2 Software Components of SAP Identity Management

SAP Identity Management consists of several components. Some of the components run on SAP NetWeaver AS for Java, for example, the Identity Management User Interface. Other components are stand-alone and are installed separately. The complete set of software units that make up SAP Identity Management are categorized as follows:



As of version 8.0 SP04 and higher, SAP Identity Management is installed using the Software Provisioning Manager 1.0 installation tool. You can use the Software Provisioning Manager 1.0 tool to install all SAP Identity Management components, except for SAP Identity Management Developer Studio client, SAP Identity Management Logon Help and SAP Identity Management Password Hook. The installation procedure for these components has not changed, that is, they must be installed manually.

Related Information

[SAP Identity Management Installation and Update Guide on Windows](#)

[SAP Identity Management Installation and Update Guide on UNIX](#)

3.2.1 License Keys

SAP Identity Management may be used by Licensee to integrate Licensee's SAP applications as part of an application-specific runtime license of SAP NetWeaver Foundation. For any other use (for example, integration with third party applications or non-SAP applications), SAP Identity Management needs to be licensed.

Related Information

[SAP Licenses](#)

3.2.2 Identity Management Core Components

The Identity Management core component consists of the Identity Management database and the Identity Management Keys Utility.

The Identity Management core components consists of the following parts:

The Identity Management Database

The Identity Management database contains different types of information:

- The identity store containing the identity data. The identity store provides a uniform view of the data, regardless of the data's original source. The data is retrieved from these various repositories, consolidated, transformed into the necessary formats, and published back to the various decentralized repositories.
- The configuration of the solution with all necessary processes, forms, and jobs that are used to process the identity data.
- The database procedures providing the business logic.

Configuration Packages

With SAP Identity Management 8.0 we introduce the configuration package concept. A configuration package is a collection of configuration information including constants, scripts, repository types, processes, forms, and jobs. Users are granted access to different packages, which allows multiple users to work on the configuration and transport separately. These configuration packages are delivered as part of the SAP Identity Management core component and imported into the SAP Identity Management database to provide a starting point for the configuration of the solution.

The Identity Management Keys Utility

This utility is used to create the encryption keys that are used to secure the connection to the database and passwords and other sensitive data in the identity store.

As of version 8.0 SP04 and higher, you install SAP Identity Management Core Components using the Software Provisioning Manager 1.0 installation tool.

Related Information

[SAP Identity Management Installation and Update Guide on Windows](#)

[SAP Identity Management Installation and Update Guide on UNIX](#)

3.2.2.1 Configuration Packages

A number of configuration packages are delivered as part of the Identity Management core components. These packages contain frameworks and connectors that can be imported to the Identity Management database to provide the basic functionality as provisioning and integration with SAP Access Control.

3.2.2.1.1 Provisioning Frameworks

In the context of Identity Management, the word framework is used to designate a set of reusable jobs, tasks, and functions that are necessary when provisioning the various system types. See an overview in the table.

Framework Overview

Framework	Description
SAP provisioning framework	Provides the set of templates to use to connect SAP systems to SAP Identity Management and to set up the jobs and tasks for provisioning the corresponding users and the corresponding assignments. The connectors for this framework are delivered as separate packages. A set of default forms for the Identity Management user interface is delivered as a separate package.

Framework	Description
Governance, Risk and Compliance (GRC) provisioning framework	The GRC provisioning framework consists of a set of processes and a configuration in the Virtual Directory Server that enables the use of SAP Access Control for risk validation before user provisioning.
SAP HCM staging area identity store HCM integration framework	This framework provides a staging area identity store and framework to use when importing identity data from an SAP HCM system. You can then work with the data in the staging area before provisioning the corresponding SAP systems.
Notification framework	The notification package contains the notification task and the notification templates that are used to send notifications from the SAP provisioning framework, approval and attestation tasks.

The SAP HCM staging area identity store supplements the SAP provisioning framework by providing functions used for the specific scenario. The GRC provisioning framework is a separate framework that is used explicitly for integration with SAP Access Control. Although it is a separate framework, it can be configured and used simultaneously with the other frameworks.

3.2.2.1.2 Connectors

There are a number of connectors for SAP and non-SAP systems available as separate packages with the SAP provisioning framework.

Note

The list of connectors shown below is subject to change as additional connectors become available. There are also connectors available for connections to SAP or non-SAP systems that have been developed by partners.

- For more information about the connector overview, see [SAP Identity Management - Connector Overview](#)
- For more information about the connector packages, see *Package Content* section in [Provisioning Framework for SAP Identity Management 8.0](#)

Connector Overview of Connectors Provided with SAP Identity Management

Connector	Applicable Product / Application	Release/Platform Prerequisites
S/4HANA Connector	SAP S/4HANA (SAP Business Suite 4 SAP HANA) on-premise system	SAP S/4HANA on-premise system 2005 or higher
SCIM Connector	SAP Cloud Identity Services - Identity Provisioning	Not applicable

Connector	Applicable Product / Application	Release/Platform Prerequisites
Identity Authentication It is formerly known as SAP Cloud Identity (SCI).	SAP Cloud Identity Services - Identity Authentication	Not applicable
SuccessFactors	SuccessFactors	Not applicable
AS Java	AS Java / J2EE Engine applications Third-party products that support SPML	AS Java / J2EE Engine Release 6.40 and higher
AS ABAP	AS ABAP applications (SU01 users), SAP HCM employee data (export to SAP Identity Management)	AS ABAP: Release 4.6 and higher SAP HCM: Release 6.0 SPS 37
Dual Stack	AS Java / J2EE Engine applications Third-party products that support SPML AS ABAP applications (SU01 users), SAP HCM employee data (export to SAP Identity Management)	SAP NetWeaver Dual Stack 6.40 and higher
AS ABAP for SAP Business Suite systems	SAP Business Suite applications (provisions SU01 users plus application-specific identity information such as business partners)	SAP Enhancement Package 4 for SAP ERP 6.0 For application-specific dependencies, see the table below
Microsoft Active Directory Microsoft Exchange	Microsoft Active Directory	Microsoft Active Directory Versions with Microsoft Windows Server 2000/2003/2008/2010 Platform: MS Windows Server 2000/2003/2008/2010 Platform: MS Windows Server 2000/2003/2008/2010 Microsoft Exchange 2007/2010
SUN One	Any LDAP directory server using the generic LDAP API Novell eDirectory SunOne Directory Special requirements for other directory servers, for example, schema modifications, on a project base	Platform: Supported platforms for the respective directory server Novell eDirectory or SunOne Directory: Any release

Connector	Applicable Product / Application	Release/Platform Prerequisites
SAP HANA Connector	SAP HANA Platform Edition	SAP HANA Platform Support Package Stack 04
Lotus Notes	IBM Lotus Notes	Lotus Domino server version 8.5.3

3.2.3 Identity Management Runtime Components

The Identity Management runtime components include the runtime engine, the dispatchers, and the SAP Identity Management Dispatcher Utility.

SAP Identity Management Dispatcher Utility

This utility is used to create the dispatcher(s). You can also start and stop the dispatcher(s) with this utility. You can use either the graphical user interface or the command line interface.

Dispatcher Engine and Runtime Engine

The dispatcher and runtime engines are responsible for processing both provisioning and synchronization tasks. The runtime components require the SAP Java Virtual Machine (SAP JVM). If the runtime components run on the same server as an SAP NetWeaver AS for Java system, then they can use the SAP JVM that is provided with the AS Java system.

Note

It is recommended to use the latest SAP JVM released version.

As of version 8.0 SP04 and higher, you install SAP Identity Management Runtime Components using the Software Provisioning Manager 1.0 installation tool.

Related Information

[SAP Identity Management Installation and Update Guide on Windows](#)

[SAP Identity Management Installation and Update Guide on UNIX](#)

3.2.4 Identity Management Developer Studio

The Identity Management Developer Studio is an Eclipse plug-in that provides the environment for developing the configuration for the identity management solution. It uses the Developer Studio service running on SAP NetWeaver AS for Java.

Related Information

[Installing the Identity Management Developer Studio](#)

3.2.5 Identity Management Developer Studio Service

The Identity Management Developer Studio service provides the connection between the Identity Management Developer Studio and the Identity Management database.

This service is deployed and configured as part of SAP NetWeaver AS for Java.

The Identity Management Developer Studio service authenticates the users in the Identity Management Developer Studio against the UME.

The connection to the Identity Management database is configured as part of this service.

As of version 8.0 SP04 and higher, you deploy SAP Identity Management Developer Studio service using the Software Provisioning Manager 1.0 installation tool.

Related Information

[SAP Identity Management Installation and Update Guide on Windows](#)

[SAP Identity Management Installation and Update Guide on UNIX](#)

3.2.6 Identity Management User Interface Components

The Identity Management user interface components are the Identity Management user interface, the Identity Management administration user interface, and the Identity Management REST API version 1.

The user interface components are deployed as one software unit on SAP NetWeaver AS for Java:

Identity Management User Interface

This is the end user interface providing functions for user registration and other self-service tasks, password reset requests, and approval of assignment requests and other changes requiring approval.

Identity Management Administration User Interface

This is the user interface for system administrators and managers responsible for monitoring and maintaining SAP Identity Management.

Identity Management REST API Version 1

This RESTful Web service consists of resources with three defined aspects: the base Uniform Resource Identifiers (URIs) to access the functionality, the Internet media type of the data supported by the Web service, and the set of operations supported by the Web service, for example, GET and POST.

As of version 8.0 SP04 and higher, you deploy SAP Identity Management User Interface using the Software Provisioning Manager 1.0 installation tool.

Related Information

[SAP Identity Management REST API Version 1](#) 

[SAP Identity Management Installation and Update Guide on Windows](#)

[SAP Identity Management Installation and Update Guide on UNIX](#)

3.2.7 Identity Management REST Interface Version 2

The Identity Management REST (Representational State Transfer) Interface Version 2 is a service API (Application Programming Interface) that supports the new Identity Management User Interface for HTML5 and other new custom-made user interfaces.

Identity Management REST Version 2 implements the Open Data Protocol (OData) in version 2.0, and supports (as does OData) both formats for representing the resources it exposes - the XML-based Atom format and the JavaScript Object Notation (JSON) format.

As of version 8.0 SP04 and higher, you deploy SAP Identity Management REST Interface Version 2 using the Software Provisioning Manager 1.0 installation tool.

Related Information

[SAP Identity Management REST Interface Version 2](#)

[SAP Identity Management Installation and Update Guide on Windows](#)

[SAP Identity Management Installation and Update Guide on UNIX](#)

3.2.8 Identity Management User Interface for HTML5

SAP Identity Management User Interface for HTML5 is a user interface based on HTML5 and JavaScript, and developed using the SAP UI Development Toolkit for HTML5 (SAPUI5). It also uses SAP Identity Management REST Interface Version 2.

SAP Identity Management User Interface for HTML5 can be used by all users to maintain their own profile information and request new roles (self service). Authorizations are grouped into business roles, again made available to end-users, who can request assignment of the business roles. SAP Identity Management User Interface for HTML5 only supports assignment requests for business roles, that is, users cannot request privilege assignments.

Managers and administrators can also use SAP Identity Management User Interface for HTML5 for role request approvals. Although privilege assignment requests from the users are not supported, the [My Approvals](#) page supports approving and declining both business role assignments and privilege assignments for managers and administrators, to support cases in which approval workflows are set up for individual privileges, triggered either by automated processes or other UIs or APIs.

SAP Identity Management User Interface for HTML5 does not support processing more than one approval request at a time. You need to process each approval request separately.

SAP Identity Management User Interface for HTML5 does not support attestations, hence on the [My Approvals](#) page are shown only approvals. The attestations are available only from the SAP Identity Management REST Interface Version 2.

SAP UI Development Toolkit for HTML5 (SAPUI5)

The SAP UI Development Toolkit for HTML5 (SAPUI5) is SAP's new enterprise-ready HTML5 rendering library for client-side UI rendering and programming. It combines the advantages of being open and flexible as well as being enterprise ready, supporting all SAP Product Standards. While Web Dynpro is best suited to heavyweight transactional applications for expert usage, SAPUI5 is designed for building lightweight consumer-grade UIs

for casual usage. It is aimed at developers at SAP and customers with web development skills (HTML, CSS3, JavaScript). SAPUI5 provides extensible controls and powerful theming but is easy to consume, based on open standards, and integrates with third-party JavaScript libraries. SAPUI5 applications run on a wide range of devices (smartphone, tablet, and desktop) and on multiple server platforms.

As of version 8.0 SP04 and higher, you deploy SAP Identity Management User Interface for HTML5 using the Software Provisioning Manager 1.0 installation tool.

Related Information

[SAP Identity Management Installation and Update Guide on Windows](#)

[SAP Identity Management Installation and Update Guide on UNIX](#)

3.2.9 Virtual Directory Server

The Virtual Directory Server acts as a single access point for clients retrieving or updating data in multiple data repositories, since it provides a uniform view of the data in real time.

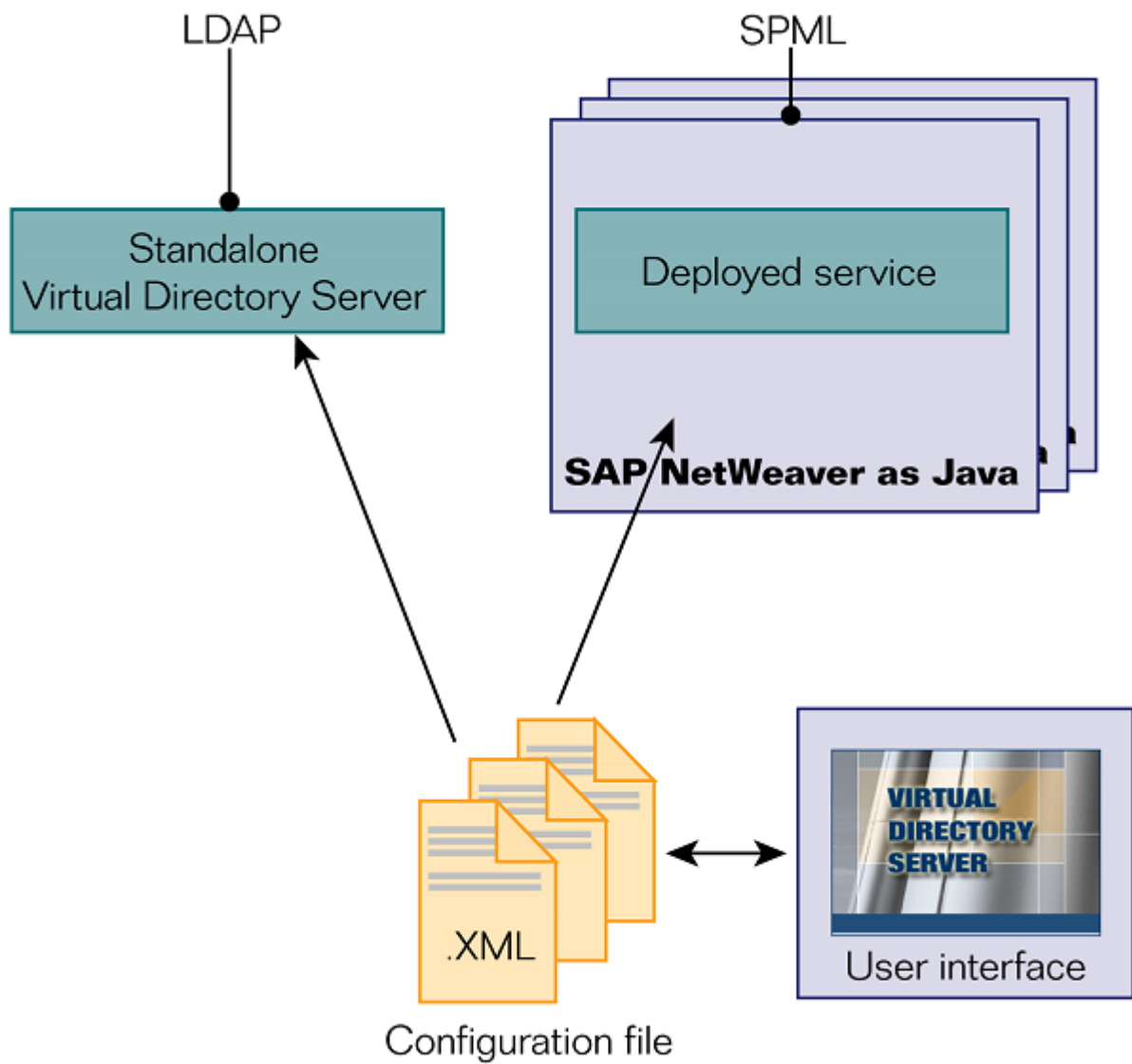
You can use it, for example, to consolidate multiple repositories into a single data source. You can then provision and perform identity management functions on the repositories using the Virtual Directory Server.

The Virtual Directory Server implements a structure called a virtual directory tree. It is a structure that organizes all managed applications so that each of them can be addressed through a unique identifier. A unique identifier, in this context, corresponds to a distinguished name in the virtual directory tree, but is mapped to a unique identifier within the application. In addition, the Virtual Directory Server has built-in connectors (and an extensible connector framework) for a variety of the applications. Most important, the Virtual Directory Server has a connector for the Identity Management database, so it can execute operations directly in the identity store.

The Virtual Directory Server provides a range of additional services such as virtualization, name space conversion, attribute and schema mapping, and attribute value modification. These services may be crucial for resolving requirements when using identity services (see the solution-wide capabilities).

Architecture Overview

The following illustration gives a high-level overview over the architecture of the Virtual Directory Server:



The user interface that is used to maintain the configuration is installed on one server, while the configurations are deployed on one or more servers running SAP NetWeaver AS for Java.

As of version 8.0 SP04 and higher, you install SAP Identity Management Virtual Directory Server using the Software Provisioning Manager 1.0 installation tool.

Related Information

[SAP Identity Management Installation and Update Guide on Windows](#)

[SAP Identity Management Installation and Update Guide on UNIX](#)

3.2.10 Identity Management Portal Content

The Identity Management Portal Content integrates SAP Identity Management with the Universal Worklist (UWL).

UWL gives users a unified and centralized way to access their work and relevant information in the portal. It collects tasks from multiple provider systems in one list for easy access to all tasks. With this architecture, you can also include tasks that originate from SAP Identity Management, for example, approvals.

As of version 8.0 SP04 and higher, you deploy SAP Identity Management Portal Content using the Software Provisioning Manager 1.0 installation tool.

Related Information

[UWL Integration Configuration Guide](#)

[SAP Identity Management Installation and Update Guide on Windows](#)

[SAP Identity Management Installation and Update Guide on UNIX](#)

3.3 Solution-Wide Capabilities

The SAP HCM staging area identity store and SPML IDS identity store supplement the SAP provisioning framework by providing functions used for the specific scenario.

The GRC framework is a separate framework that is used explicitly for integration with SAP Access Control. Although it is a separate framework, it can be configured and used simultaneously with the other frameworks.

Additional Capabilities

Capability	Description	More Information
Identity Services	The SAP Identity Management Identity Services provide Web service access to identity information stored in an identity store in the Identity Center or some other application that can be accessed from the Virtual Directory Server. The identity services are Web services that are created and configured on the Virtual Directory Server and deployed on the AS Java.	Identity Services - Architectural Overview Identity Services: Configuration Guide

Capability	Description	More Information
Reporting (with SAP Business Warehouse)	You can use SAP Business Warehouse for reporting on identities. This option uses a BW connector on the Virtual Directory Server for transferring the data to the BW system.	Identity Reporting Using SAP Business Warehouse
Custom Implementation	You may need to extend the capabilities of SAP Identity Management to meet your own needs. For example, you may want to provision additional attributes, or you may want to trigger specific events when an identity is created or modified. For ABAP-based SAP systems, you can implement the Business Add-In (BAdI) interface IF_BADI_EXTEND_IDENTITY. This interface is available for use with the enhanced SAP Business Suite use case for the SAP provisioning framework.	SAP Identity Management Extension Framework Implementation Guide Extending the SAP Provisioning Framework

4 Overall Implementation Sequence

The overall implementation sequence is set up in three main phases:

1. Planning phase
2. Implementation and test
3. Go-live

Planning phase

The first phase of the implementation sequence for SAP Identity Management is the planning phase. In this phase, you should:

- Analyze your platform and system requirements and determine your system landscape. In addition to taking into account system requirements like security, scalability, and performance, we recommend using a staged approach. Do the initial implementation in a development system and move the configuration into a quality system for testing, and finally into the production system.
- Take organizational steps to define the roles and responsibilities needed for the implementation phase.
- Define a role model that specifies how the various roles and privileges are represented and provisioned to the various target systems.

→ Tip

We recommend you take the opportunity to clean up superfluous or outdated roles and privileges in your system. Consider using business roles to consolidate the authorization information into a central point of administration.

- Identify data ownership. This involves determining the originating and target systems for all objects and their attributes that are to be handled in the identity management landscape. This is the basis for configuring attribute mappings in the initial load jobs, update jobs, and provisioning tasks. This also provides you with an overview of which connectors and frameworks you require.
- Determine customer-specific requirements for workflows, approval tasks, reporting, or extending the frameworks that are available out-of-the box.
- Be aware of the following [factors that influence performance](#) before proceeding with the next phase.

Implementation phase

The implementation phase could include the following steps:

1. Download and install the various components in the development environment.
2. Perform the initial configuration.
3. Familiarize yourself with the product at a technical level.

This reduces errors when proceeding with the implementation.

4. Import the packages containing the individual frameworks and connectors required for your system landscape.
5. Configure and run the initial loads.
After this step, the identity data is collected in the identity store.
6. Clean up the data in the identity store.
7. Set up additional processes, for example, approvals, self-services, reporting, or custom jobs.
8. Implement an authorization concept for using and working with SAP Identity Management. This includes setting up access to the user interfaces as well as specifying attribute owners or setting up access control for specific processes and forms.
9. Install and configure all software components in the QA environment.
10. Test the complete implementation, normally in a separate QA environment.

Go-live

Once all tests are successful, move the implementation to the production environment.

1. Install and configure all software components in the production environment.
2. Copy the relevant packages to the production environment and verify that it works as expected.

Related Information

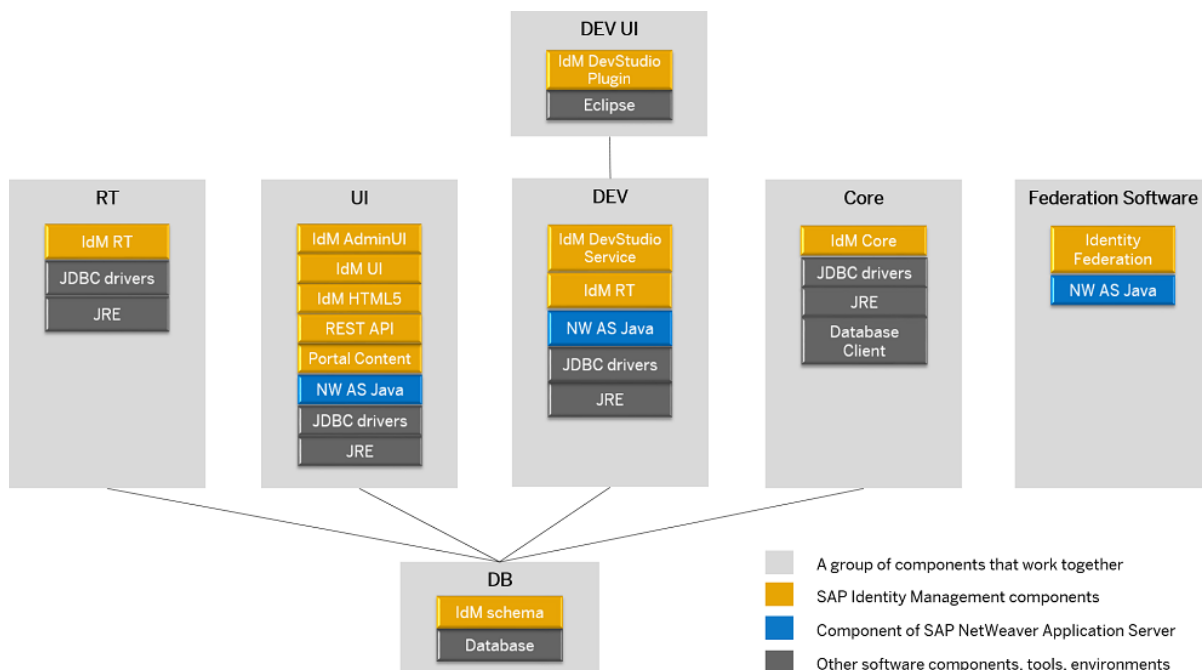
[Installation Overview \[page 25\]](#)

4.1 Installation Overview

The software components can be installed on the same server or on several servers, depending on the requirements and purpose of the installation. For a smaller development environment, all components could be installed on the same server.

In a production environment, the components are normally divided between several servers prepared for high availability and high performance:

- The database server must be clustered to ensure high availability of the data. See [High Availability](#)
- The servers with SAP NetWeaver AS for Java for the Identity Management User Interface must be clustered, to ensure high availability. Load balancing is handled by SAP NetWeaver AS for Java.
- The servers with the runtime components are duplicated by setting up two or more servers with identical configurations. This will ensure high availability and load sharing of the processing. The runtime components can also be distributed to the servers with SAP NetWeaver AS for Java.



All components of SAP Identity Management 8.0 are available for download from the SAP Software Download Center on the SAP Support Portal. Install the prerequisites and the components in accordance with the diagram above and the installation guides accompanying each of the components.

Related Information

[Software Components of SAP Identity Management \[page 11\]](#)

[SAP NetWeaver Identity Management 7.1/7.2](#)

5 SAP Identity Management Scenarios

This section describes some common implementation scenarios for SAP Identity Management.

Related Information

[Provisioning for SAP or Non-SAP Systems \[page 27\]](#)

[Integration with SAP HCM \[page 29\]](#)

[Enhanced SAP Business Suite Integration \[page 31\]](#)

[Integration with SAP Access Control \[page 34\]](#)

[Logon Help \[page 36\]](#)

[Identity Federation \[page 39\]](#)

5.1 Provisioning for SAP or Non-SAP Systems

Description

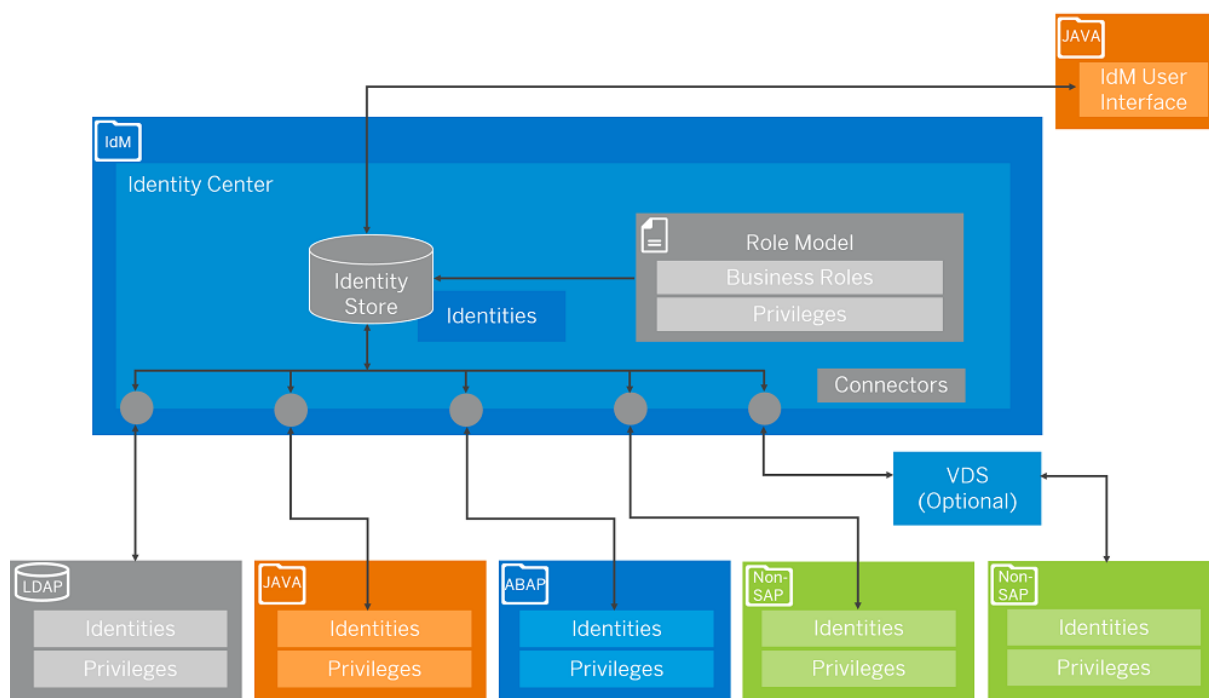
You can use SAP Identity Management for processing identity information in a variety of ways, depending on your system landscape. You can use it in homogeneous or heterogeneous landscapes, either with or without SAP systems. The identity store is the central storage location for the identity data, and when changes occur to identity-related data, including roles, privileges, and the corresponding assignments, the identity-related information is provisioned to the appropriate target systems.

Technical System Landscape

The figure below shows the basic system landscape to use for this scenario. The Identity Center is the central component where you set up the provisioning tasks and jobs, as well as the connectivity to the target systems. The Identity Center also hosts the role model and the data ownership model that are used to determine which identity and privilege assignments and which attribute values are provisioned to which systems.

You can use the Virtual Directory Server to consolidate systems (as appropriate) and then connect the Virtual Directory Server to the Identity Center. The Identity Management User Interface, where you make changes to the identities and other identity-related information, runs on the AS Java.

See the figure below.



Software Units

The following components are used in this scenario:

- Identity Center
- Virtual Directory Server (optional)
- Identity Management User Interface

The following connectors are used in this scenario:

- SPML connector (for AS Java target systems, or non-SAP systems that use SPML)
- AS ABAP connector (for AS ABAP target systems)
- LDAP connector (for directory servers)
- Additional connectors (as appropriate for the target systems)

In addition, the SAP provisioning framework is used when connecting to SAP systems.

Implementation Sequence

For an overview of the implementation sequence, see the Overall Implementation Sequence.

The following documents provide more information about provisioning to SAP or non-SAP systems.

Related Information

[Overall Implementation Sequence \[page 24\]](#)

[Connecting SAP Systems](#)

[Connecting Non-SAP Systems](#)

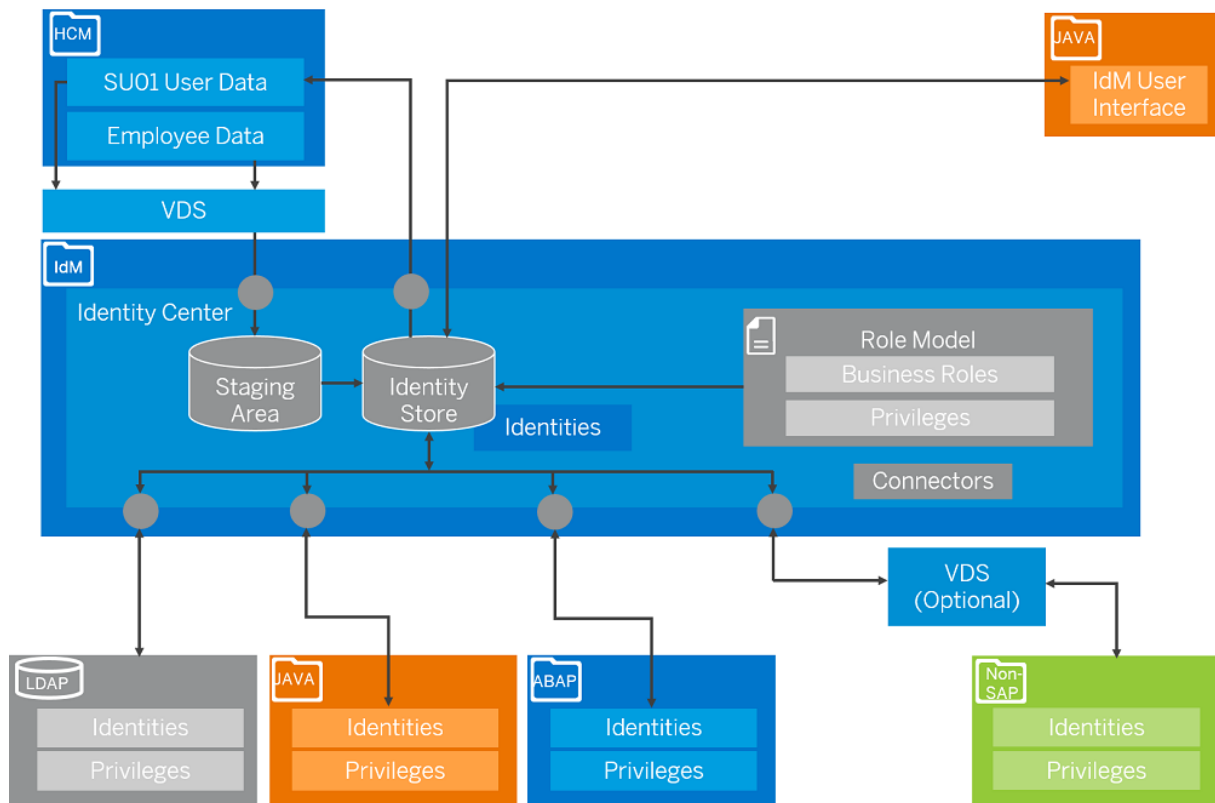
5.2 Integration with SAP HCM

Description

In many cases, the primary source for identity information (employee master data) is the SAP HCM system. When integrating SAP Identity Management with SAP HCM, identities are replicated to the Identity Center after they are created in the SAP HCM system. Based on the role model that is set up in the Identity Center, SAP Identity Management determines the user/role or user/group assignments that are provisioned to the various target systems.

Technical System Landscape

The data transfer from the SAP HCM system to SAP Identity Management takes place using the Virtual Directory Server. The Virtual Directory Server exposes an LDAP interface towards the identity store, allowing the SAP HCM system to write to the identity store using the LDAP capabilities of the AS ABAP. As in the basic scenario for provisioning to SAP or non-SAP systems, the identities and privilege assignments are provisioned to the target systems based on the role model that is set up in the Identity Center. See the figure below.



Software Units

The following components are used in this scenario:

- Identity Center
- Virtual Directory Server
- Identity Management User Interface

The following connectors are used in this scenario:

- SPML connector (for AS Java target systems, or non-SAP systems that use SPML)
- AS ABAP connector (or the AS ABAP for SAP Business Suite connector, if used in combination with the enhanced SAP Business Suite integration scenario)
- LDAP connector (for directory servers)
- Additional connectors (as appropriate for the target systems)

In addition, the SAP provisioning framework and the SAP HCM staging area identity store are used in this scenario.

Implementation Sequence

For an overview of the implementation sequence, see the Overall Implementation Sequence.

The following documents provide more information about integration with SAP HCM systems.

Related Information

[Setting Up an SAP HCM System](#)

5.3 Enhanced SAP Business Suite Integration

Description

In addition to SAP HCM, you can integrate many applications from the SAP Business Suite into the SAP Identity Management landscape. In this case, application-specific processing such as the creation of a business partner is performed in addition to the provisioning of standard AS Java or AS ABAP identities (SU01 users) and their corresponding assignments. The corresponding connector is provided with the SAP provisioning framework.

Note

For many of the SAP Business Suite systems, for example, SAP CRM or SAP SRM, a central person is created and used to link an identity to his or her business partners. When an identity is created and provisioned with SAP Identity Management, this central person and the corresponding business partner is also created in the SAP Business Suite system.

Another enhancement available in this scenario is that certain communication data for the employee can be provisioned back to the SAP HCM system. This is not possible in the standard SAP HCM scenario. The table below shows the applications that are supported by the AS ABAP for SAP Business Suite connector, additional application-specific release prerequisites, if applicable, and the feature provided for the application.

SAP Business Suite Systems and Features Supported with Enhanced Business Suite Integration

SAP Business Suite Application	Features	Prerequisites
SAP Human Capital Management	Sending of employee-related data from SAP HCM to SAP Identity Management. Transfer of identity data, including communication data, from SAP Identity Management to SAP HCM.	SAP HCM application component Personnel Administration as of SAP Enhancement Package 4 for SAP ERP 6.0

SAP Business Suite Application	Features	Prerequisites
SAP ERP Financials (Auditing)	A user with the role SAP_PLM_AUDITOR will also receive authorizations for the transactions Audit Management and Audit Monitor, as soon as the user and authorization distribution has been completed.	CA-AUD (auditing) of SAP ERP cross-application components as of SAP Enhancement Package 4 for SAP ERP 6.0
SAP ERP Financials (Accounting)	A new SAP Financials user automatically receives access to all of the functions for the corresponding company code that apply to his or her responsibility.	A new SAP Financials user automatically receives access to all of the functions for the corresponding company code that apply to his or her responsibility.
SAP Transportation Management (SAP TM)	The combination of a user account, a business partner, and a central person is created automatically.	SAP HCM application component Personnel Administration as of SAP Enhancement Package 4 for SAP ERP 6.0 (optional) SAP TM 7.0 or higher
SAP Extended Warehouse Management (EWM)	The combination of a user account, a business partner, and a central person is created automatically.	SAP HCM application component Personnel Administration as of SAP Enhancement Package 4 for SAP ERP 6.0 SAP EWM 7.0 or higher with labor management activated
SAP Supply Network Collaboration	Triggers automatic generation of users and business partners for SAP SNC.	SAP HCM application component Personnel Administration as of SAP Enhancement Package 4 for SAP ERP 6.0 SAP SNC 7.0 or higher
SAP Service Parts Planning (SPP)	Triggers automatic generation of users and business partners for SAP SPP.	SAP HCM application component Personnel Administration as of SAP Enhancement Package 4 for SAP ERP 6.0 (for the creation of users and business partners for new employees)
SAP Product Lifecycle Management	Users are created in PLM, based on employee data from SAP HCM.	SAP HCM application component Personnel Administration as of SAP Enhancement Package 4 for SAP ERP 6.0 The PLM Web User Interface (PLM Web UI) is activated.
SAP Portfolio and Project Management	The combination of a user account, a business partner, and a central person is created automatically.	SAP HCM application component Personnel Administration as of SAP Enhancement Package 4 for SAP ERP 6.0
SAP Customer Relationship Management (SAP CRM)	SAP Customer Relationship Management (SAP CRM)	SAP CRM 7.0

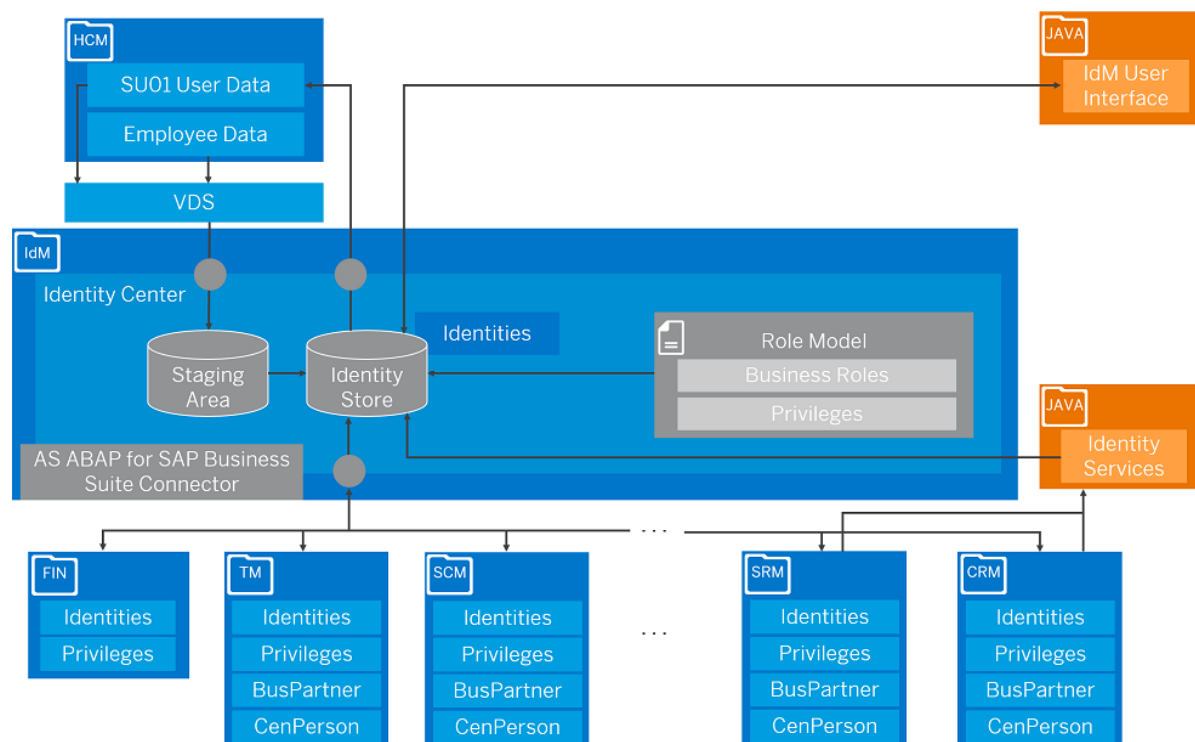
SAP Business Suite Application	Features	Prerequisites
SAP Supplier Relationship Management (SAP SRM)	The combination of a user account, a business partner, and a central person is created automatically.	SAP ERP HCM as of SAP Enhancement Package 4 for SAP ERP 6.0 SAP SRM 7.0

Technical System Landscape

The system landscape to use for this scenario is similar to the landscape for the other scenarios that involve SAP systems. Typically, the SAP HCM system is set up as the starting point for maintaining identity data, which is then provisioned to the target systems. The difference in this scenario is that the AS ABAP for SAP Business Suite connector is used to connect to the corresponding SAP Business Suite systems instead of the AS ABAP connector. This allows for the additional application-specific processing of the identity information.

In addition, certain SAP Business Suite applications (for example, by SAP CRM or SAP SRM) send identity-related information to SAP Identity Management using identity services, which run on an AS Java.

See the figure below.



Software Units

The following components are used in this scenario:

- Identity Center
- Virtual Directory Server (assuming the SAP HCM is included in the system landscape)
- Identity Management User Interface

The following connectors are used in this scenario:

- SPML connector (for AS Java target systems, or non-SAP systems that use SPML)
- AS ABAP for SAP Business Suite connector (for SAP Business Suite target systems)
- LDAP connector (for directory servers)
- Additional connectors (as appropriate for the target systems)

The following frameworks are used in this scenario:

- SAP provisioning framework
- SAP HCM staging area identity store
- SPML IDS identity store (for SAP CRM and SAP SRM applications)

Implementation Sequence

For an overview of the implementation sequence, see the Overall Implementation Sequence.

The following documents provide more information about enhanced SAP Business Suite Integration.

5.4 Integration with SAP Access Control

Description

The integration with SAP Access Control consists of a set of tasks in the Identity Center and a configuration in the Virtual Directory Server that enables the use of SAP Access Control for risk validation before user provisioning. Using this solution, SAP Identity Management can execute provisioning to multiple target systems that are controlled by SAP Access Control to ensure compliance in accordance with the rules implemented here.

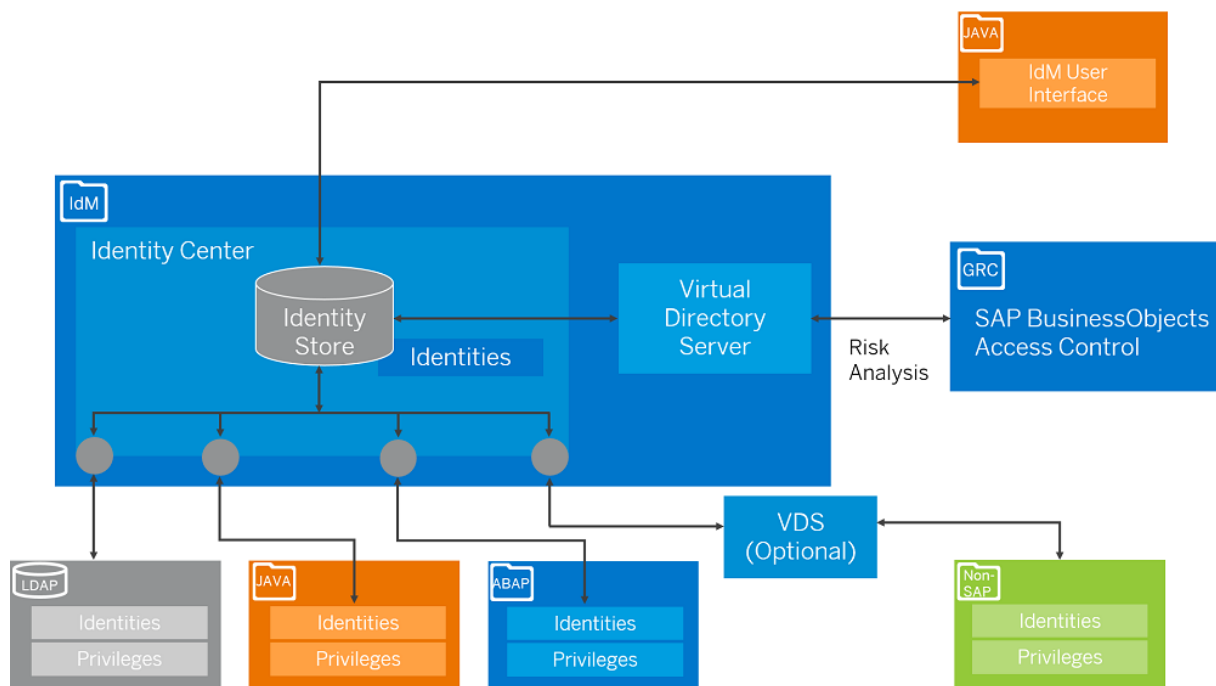
When business requirements imply compliance and segregation of duties checks, SAP Identity Management performs risk validation on SAP Access Control before assigning permissions.

Technical System Landscape

There are two landscape configuration scenarios for the integration:

- Centralized provisioning
We recommend centralized provisioning as the default solution. This is a scenario where SAP Identity Management is the only provisioning system, responsible for provisioning both the assignments requiring and not requiring compliance checks to the systems (both SAP and non-SAP). SAP Identity Management uses SAP Access Control to execute risk analysis.
- Distributed provisioning
This solution is recommended for use only in exceptional cases. The provisioning is performed both by SAP Identity Management and SAP Access Control.

The figure below shows an overview of the system landscape when using centralized provisioning.



Software Units

The following components are used in this scenario:

- Identity Center
- Virtual Directory Server
- Identity Management User Interface

In addition to the connectors for identity provisioning to the target systems, the SAP Access Control (GRC) connector is needed in this scenario. In addition to the SAP provisioning framework, the GRC framework is needed in this scenario.

Implementation Sequence

If SAP Identity Management is to perform the provisioning tasks, set up provisioning to the target systems based on the overall implementation sequence. In addition, set up the integration with SAP Access Control as follows:

1. Create the corresponding configuration on the Virtual Directory Server.
2. Extend the Identity Center identity store schema.
3. Import the SAP GRC provisioning framework and corresponding service jobs into the Identity Center.
4. Adjust the Identity Center and Virtual Directory Server configurations.
5. Initialize the process by running the initial load jobs.

For more information about SAP Access Control integration, including detailed information about the implementation steps, see the documents listed in the table below.

Related Information

[Integration with SAP Access Control 10.0 or higher using the GRC 10.0 or higher Provisioning Framework Version 2](#)

5.5 Logon Help

SAP Identity Management Logon Help is a client application for Microsoft Windows.

→ Recommendation

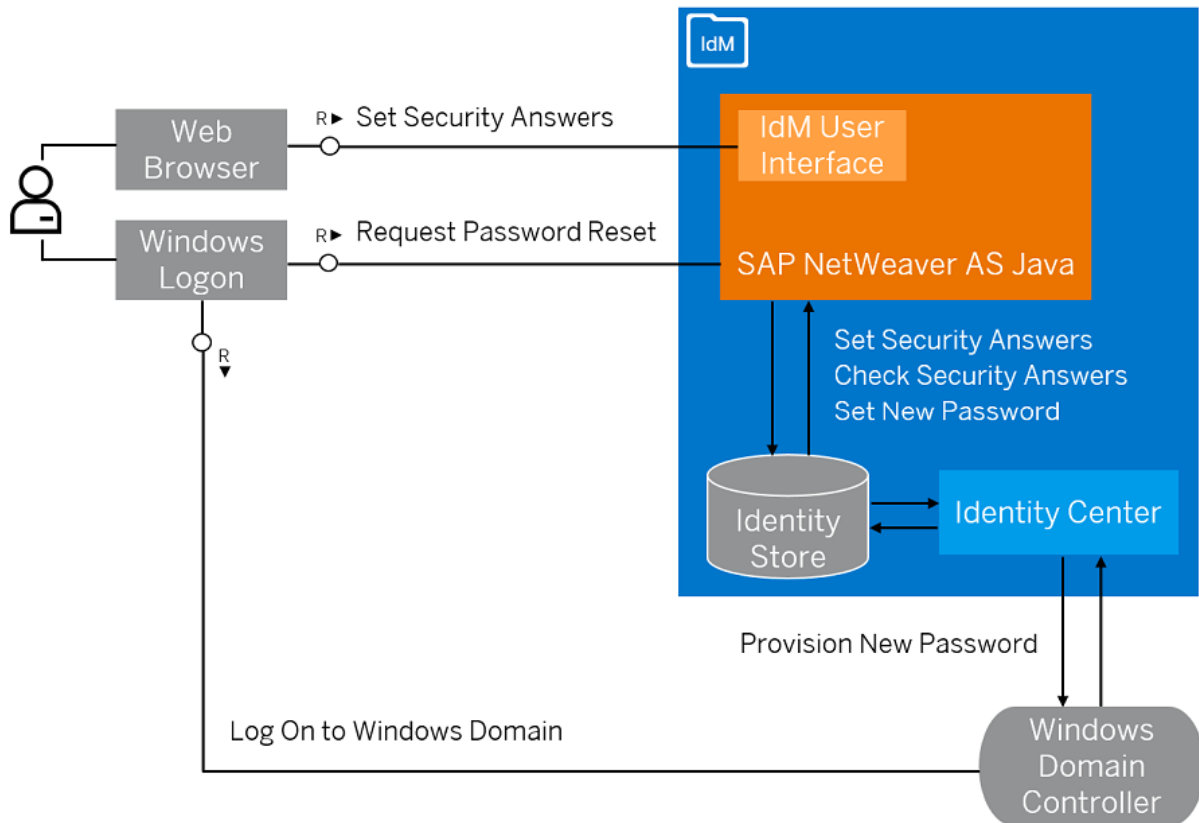
When installing SAP Identity Management 8.0, we recommend that you no longer install the Logon Help component. This is due to the fact that Logon Help is supported for Windows 7 which is out of maintenance. Existing customers, who are already using Logon Help component with Windows 7, will still be supported. For new installations and scenarios customers can use the Windows build-in alternative.

Description

Logon Help enables users in a Microsoft Windows domain to reset their passwords from the Microsoft Windows logon screen by answering a set of security questions. SAP Identity Management Identity Center checks that the answers are correct and provisions the new password into the Microsoft Windows domain. Once complete, Logon Help enables the user to log on to the Microsoft Windows domain. Logon Help empowers users to reset their own passwords without having to resort to local call centers for a password reset.

Technical System Landscape

Logon Help communicates with the Identity Center to get the required information to enable password resets for users of Microsoft Windows.



The figure above illustrates a user logging on to the SAP Identity Management user interface to set security questions and answers, which the Identity Center saves in the identity store. When a user forgets his or her password, the user starts the Logon Help client from the Microsoft Windows logon screen. Logon Help retrieves the security questions from the identity store through the Identity Center. The user then provides the required answers and a new password. Logon Help passes the answers to the Identity Center, which checks them against the hash values of the answers stored in the identity store. If they match, the Identity Center resets the password and provisions the new password to the network. Logon Help then polls the domain controller until the new password is available and logs the user on.

Software Units

The system landscape required for SAP Identity Management Logon Help assumes the following components:

- **Microsoft Windows domain**
You have a network of workstations in one or more Microsoft Windows domains with your users in an active directory server (ADS).

- SAP Identity Management Identity Center 7.2 SP 06 or higher

Implementation Sequence

The implementation sequence for Logon Help requires the following steps in addition to the overall implementation sequence.

1. Configure the user management engine (UME) of the SAP NetWeaver AS Java to use the ADS as the data source.
The mapping of the UME to the ADS enables your Microsoft Windows users to log on to the SAP Identity Management user interfaces so they can set their security questions.
2. For the Identity Center to read data from the ADS into its identity store, the following prerequisites must be met:
 - The Identity Center is configured to use the ADS as its data source.
 - To provision passwords to the Active Directory, you need to set up an SSL connection between the Identity Center and the Active Directory. At least indicate the SSL port of the Active Directory in the repository constants. For more information, see *Repository Constants for Active Directory for Provisioning Framework* in *SAP Identity Management Configuration Guide*.
 - On the AS Java, the administrator has assigned the users of the Active Directory at least the UME action to access the Identity Center UI.

Then execute the steps below:

- Under **Enterprise People** > **Password Policy** > **...**, enable **Password Provisioning**.
 - Configure the self-service password reset feature. This also includes defining a UI task to enable the business users to enter the responses to the security questions. This is only possible after making the task available to all users.
 - Use the **Set Password On AD** task as part of the provisioning framework.
3. Install and configure SAP Identity Management Logon Help on the client computers.

The following documents provide more information about Logon Help for SAP Identity Management.

Related Information

[Repository Constants for Active Directory for Provisioning Framework](#)
[Setting Up a Microsoft Active Directory and Microsoft Exchange](#)

5.6 Identity Federation

Description

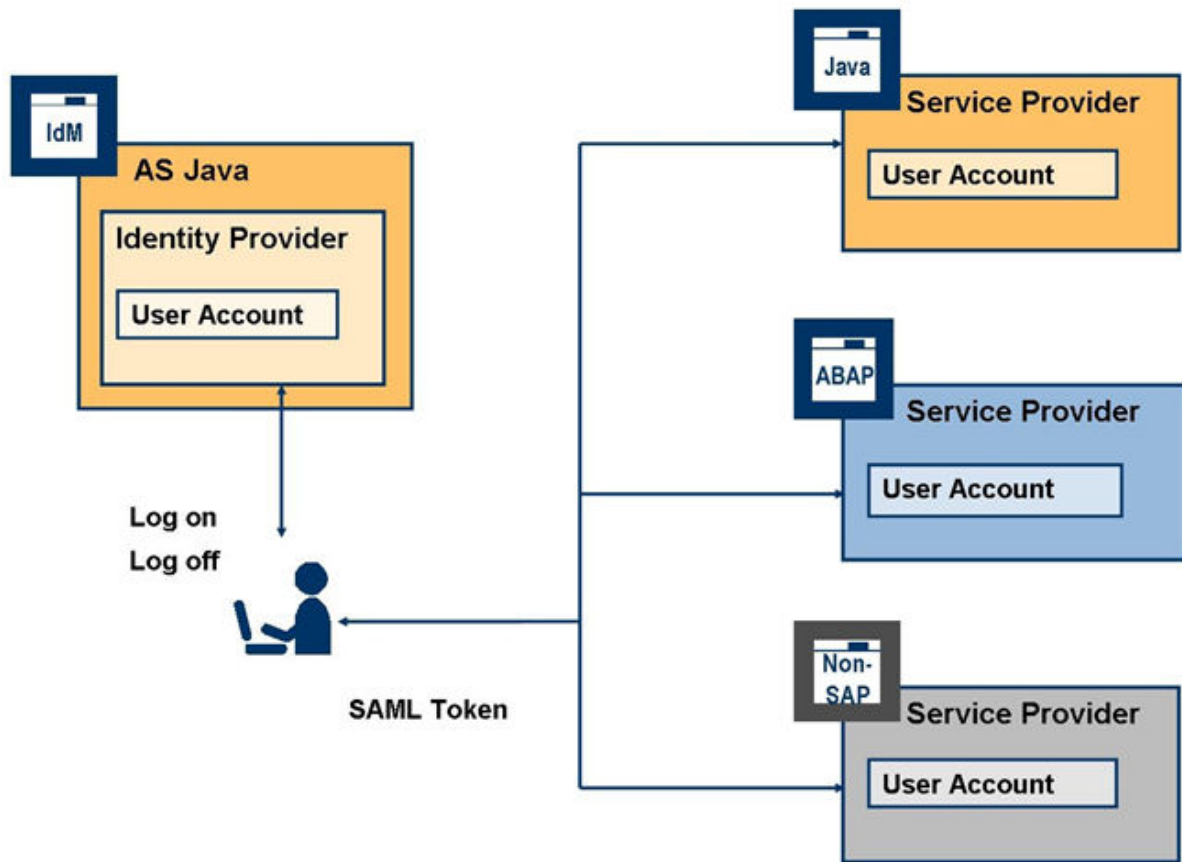
Identity federation provides the means to share identity information across company boundaries. To share information about a user, partners must be able to identify the user, even though they may use different identifiers for the same user. The name identifier (name ID) is the means to establish a common identifier. Once the name ID has been established, the user is said to have a federated identity. Identity federation enables SSO for Web-based access and Web services across domains, such as between companies. SAP's solution relies on standards for interoperability between SAP and non-SAP systems. For Web-based access, identity federation uses an identity provider that supports SAML 2.0. SAML 2.0 also enables Single Log-Out (SLO). You can also use identity federation to transport profile attributes to create or update temporary or permanent users between systems. You can even transport authorization attributes enabling you to change user authorizations in a target system. For Web services, identity federation uses a security token service (STS) that supports WS-Trust 1.3. The STS supports a number of authentication methods from a Web service consumer and can convert these tokens into a security token that a Web service provider can use. The STS supports X.509, SAML 1.1, and SAML 2.0 tokens. Like SAML 2.0 for Web-based access, the SAML 2.0 assertion can transport profile and authorization attributes to the target Web service provider.

Technical System Landscape

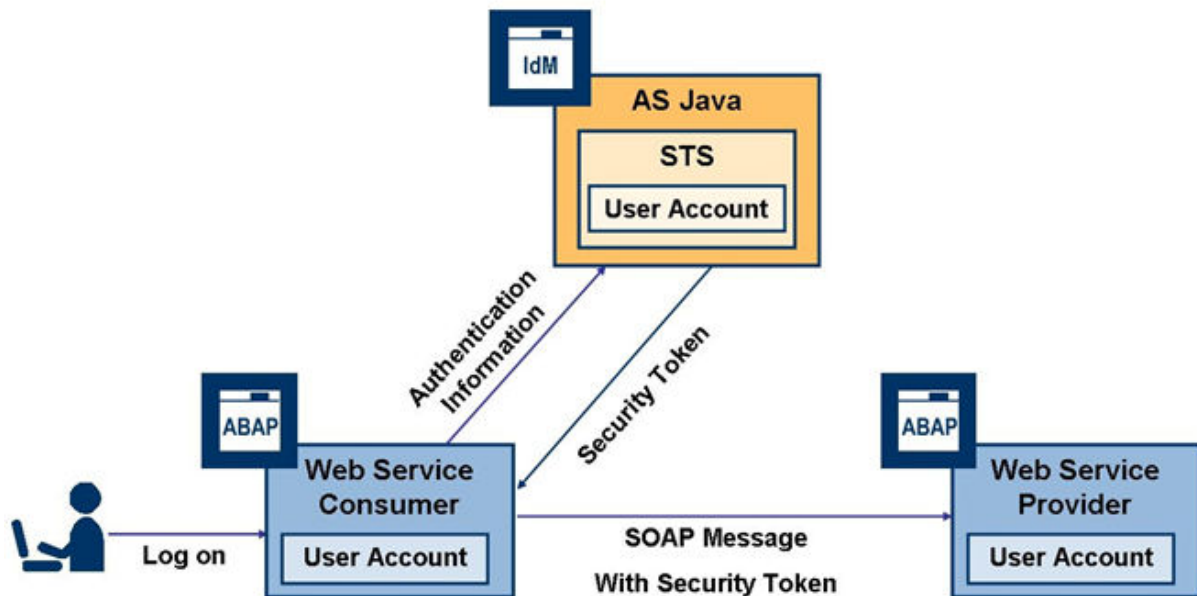
The figures below show an overview of example system landscapes when using federation.

→ Tip

Protect all communication between systems with Secure Sockets Layer (SSL) especially those that carry messages that are not already encrypted.



Identity federation for Web-based access relies on an identity provider that links a local account to a number of user accounts on service providers with a name ID. When a user logs on to the service provider, the service provider only needs the name ID to log the user on to the local account.



Identity federation for Web services relies on an STS to provide a security token to a Web service consumer. Before the STS can issue a security token, it needs authentication credentials for the local user of the STS. The STS provides the name ID (or subject for X.509 tokens) that the Web service consumer uses to authenticate the user at the Web service provider. The figure above uses a Web service consumer and Web service provider of an AS ABAP, but the solution is not limited to the AS ABAP or even SAP consumers and providers.

Software Units

For Web-based access, the primary component used for federation is the identity provider, which runs on the AS Java. The target systems that are to be included in the federation scenario also need to be active service providers.

For Web services, the primary component used for federation is the STS, which runs on the AS Java. The target systems that are to be included in the federation scenario also need to be active Web service consumers and Web service providers.

Implementation Sequence

The implementation sequence for the federation scenarios differs from the overall implementation sequence.

- Web-Based Access
 1. Download and install the federation software.
 2. Configure the identity provider.
 3. Enable the identity provider.
 4. Configure the types of protocol bindings to support.

5. Identify and configure the trusted service providers.
- Web Services
 1. Download and install the federation software.
 2. Configure the STS.
 3. Enable the STS.
 4. Select the authentication types for Web services.
 5. Trust the Web service providers.
 6. Identify and configure the trusted Web service providers.
 7. Identify and configure the Web service consumers.

For more information about identity federation, including detailed information about the implementation steps, see the following documents:

Related Information

[Identity Provider for SAP Single Sign-On and SAP Identity Management](#)
[Security Token Service for SAP Single Sign-On and SAP Identity Management](#)

6 Appendix

The following list shows all documents mentioned in this Master Guide.

Related Information

[SAP Identity Management Installation and Update Guide on Windows](#)
[SAP Identity Management Installation and Update Guide on UNIX](#)
[SAP Identity Management Security Guide](#)
[SAP Identity Management Solution Operation Guide](#)
[SAP Identity Management - Connector Overview](#) 
[Sizing Guide for SAP NetWeaver Identity Management 7.1/7.2](#)
[SAP Identity Management Configuration Guide](#)
[SAP Identity Management Developer Studio](#)
[Provisioning Framework for SAP Identity Management 8.0](#)
[Identity Reporting Using SAP Business Warehouse](#)
[SAP Identity Management Extension Framework Implementation Guide](#)
[Identity Provider for SAP Single Sign-On and SAP Identity Management](#)
[Security Token Service for SAP Single Sign-On and SAP Identity Management](#)
[SAP Identity Management REST Interface Version 2](#)
[Integration with SAP Access Control 10.0 or higher using the GRC 10.0 or higher Provisioning Framework Version 2](#)
[Logon Help for SAP Identity Management Implementation Guide](#)

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